Behavioural-to-Video-Ephys Temporal Alignment and Neural Circuit Characterisation

Full Project Report: Video Sync, Feature-Based Alignment, and Spike-Sorted Anchor Alignment

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Neuropixels 1.0, SpikeGLX, Kilosort4

Head-fixed Dynamic Target Task (joystick pull for water reward)

Phase 0	Video–behaviour alignment (PC1 / template ensemble)
Phase 1	Continuous-feature ephys–behaviour alignment (440+ features)
Phase 2	Spike-sorted anchor-unit alignment and circuit analysis

Sessions: Ferrero R1–R4 (imec0/1), Milka R1–R4 (imec0/1)
Recorded August 2023 Analysed 2025–2026

Contents

1 Introduction and Project Scope

This report consolidates three sequential phases of a single alignment problem: relating events in a behavioural log (a head-fixed joystick-pull task delivering water reward) to two independently clocked recording systems — a behaviour-tracking video camera and a Neuropixels 1.0 electrophysiology probe. Each phase was driven by the failure or partial success of the one before it, and each used a conceptually different signal as its anchor.

- **Phase 0 (Video alignment).** An ensemble video-tracking method (PC1 trace + template matching + spatial stability) was used to detect a physical event (joystick/water-port motion) in raw video and match it to behaviour-log timestamps. This phase validated that confidence-weighted ensemble detection plus a hyperparameter sweep could produce a usable, if imperfect, video–behaviour warp.
- **Phase 1 (Continuous-feature ephys alignment).** With no functional hardware sync channel in the Neuropixels recordings, over 440 continuous electrophysiological features (solenoid artefact proxies, MUA envelope, LFP band power, phase-amplitude coupling, reward-modulation index) were engineered and audited as candidate alignment anchors. This phase concluded that no continuous feature was specific enough to reward delivery to support precise per-trial alignment.
- **Phase 2 (Spike-sorted anchor alignment).** Individual well-isolated units from the Kilosort4 spike-sorting output were used as biological clocks: a single reward-responsive neuron, matched to each behavioural reward event, supports a linear clock warp with sub-millisecond to $\sim$20 ms precision. This phase succeeded and is used as the foundation for downstream neural circuit analysis.

The remainder of this report covers each phase in turn (Sections 2–4), followed by a cross-phase discussion (Section 5) and conclusion (Section 6).

2 Phase 0: Video–Behaviour Alignment

2.1 Motivation

Before any ephys-specific work began, a video camera recording of each session needed to be aligned to the behaviour log independently, both as a sanity check on the behavioural pipeline and as a potential secondary validation signal for later ephys alignment work. Session 4_Milka R1 (2023-08-15) was used as the primary test case.

2.2 Method

`video_alignment_run_ready.py` processed the full video at a frame step of 10 (180,745 frames sampled at $\sim$3–4 hours runtime) and extracted, per frame, an ROI-based PC1 trace and template-match score intended to detect a stereotyped visual event (joystick pull / water-port motion) corresponding to each behavioural reward.

Detection confidence was computed from an ensemble of three cues: PC1 peak magnitude, template-match score, and spatial coordinate stability (whether the matched ROI location was consistent across nearby frames). A confusion-matrix-free continuous confidence score, rather than a binary detector, was used because the relationship between any single cue and ground truth was empirically unpredictable across trials.

2.2.1 Hyperparameter Sweep

A grid search over `min_conf` $\in \{0.2, 0.3, 0.35, 0.4, 0.5\}$ $\times$ `pc1_weight` $\in \{0.3, 0.45, 0.6\}$ (15 configurations, 128 valid trial pairs per configuration) was run to select the ensemble-weighting parameters. The rationale, as noted during the original run, was that higher confidence thresholds trade false positives for false negatives in a way that is difficult to predict analytically given the unpredictable relative reliability of PC1 versus template matching across trials — so the sweep was evaluated empirically rather than fixed a priori.

Table 1: Video-alignment hyperparameter sweep, Milka R1 (selected rows).

min_conf	pc1_weight	stretch	MAE (s)	RMSE (s)
0.3	0.6	1.0023	1.064	151.08
0.35	0.3	1.0015	2.686	151.18
0.2	0.45	1.0013	3.672	151.12
0.4	0.6	1.0019	1.593	151.14
0.3	0.45	1.0030	−0.236	150.95
0.2	0.3	1.0021	1.150	151.07
0.5	0.3	1.0020	1.436	151.06

The best configuration by MAE was `min_conf` = 0.3, `pc1_weight` = 0.6 (MAE = 3.09 s, RMSE = 3.68 s after outlier handling in the final fit; see Table 2). All 15 configurations converged to a very similar stretch ($\approx$ 1.001–1.003) and a near-identical RMSE plateau around 151 s, reflecting that the sweep was dominated by a fixed offset-bound limitation described below rather than by genuine sensitivity to the ensemble weights.

2.3 Results

Table 2: Video–behaviour warp summary, Milka R1.

Metric	Value
Warp stretch	1.0021
Warp offset	1.15 s
Clock drift (total)	∼4.8 s over 2300 s
Trials aligned	138 / 138 (100%)
Median confidence	∼0.80
Min / max confidence	0.51 (trial 129) / 1.00 (trials 1, 8, 26, 41, 113, 124)

The fitted linear relationship was $t^{\text{video}} = 1.0021\, t^{\text{bhv}} + 1.15\text{s}$ (Figure 1, top), with the video–behaviour offset growing approximately linearly across the session at a drift rate of 33.70 ms/trial (Figure 1, centre) — consistent with a small constant clock-rate mismatch between the video capture clock and the behaviour PC clock, accumulating to ∼4.8 s of total drift over the ∼2300 s analysed window.

video_warp_diagnostics.png

Figure 1: Video–behaviour alignment diagnostics, Milka R1. **Top:** Behaviour time vs. video time for 138 matched trials, coloured by detection confidence; red line: fitted warp ($t^{\mathrm{video}} = 1.0021\, t^{\mathrm{bhv}} + 1.15\mathrm{s}$). **Centre:** Clock offset per trial, showing a clear linear drift trend (33.70 ms/trial). **Bottom:** Confidence vs. |offset| magnitude — a weak negative trend (slope -0.0134) indicates that larger clock offsets are mildly associated with lower detection confidence, as expected if confidence partly reflects matching difficulty late in the session.

A representative sample of the per-trial alignment table is given in Table 3.

Table 3: Per-trial video alignment (sampled rows), Milka R1.

Trial	Bhv time (s)	Video time (s)	Offset (s)	Confidence
0	1062.37	1065.75	3.38	0.883
1	1164.07	1167.67	3.60	1.000
20	1406.99	1411.10	4.11	0.605
50	1905.62	1910.99	5.37	0.831
90	2527.53	2534.00	6.47	0.755
120	3068.21	3075.79	7.58	0.867
137	3360.44	3368.64	8.20	0.649

A separate diagnostic, the PC1 z-score trace overlaid with logged reward times, confirmed the qualitative validity of the video signal independent of the fitted warp: clear drops in the PC1 trace align with reward times once the task is underway, and the pre-task quiescent period (before $\sim$100 s) is visually distinct from the active task period (Figure 2).

video_pc1_trace.png

Figure 2: PC1 trace vs. logged reward times, Milka R1. **Top:** Full-session PC1 z-score trace (blue) with reward times overlaid (red vertical lines). The pre-task quiescent period before $\sim$100 s is visually distinct from the active period. **Bottom:** Zoomed view (1050–1350 s) showing individual PC1 drops aligning with reward delivery.

2.4 Known Issues and Carry-Forward

Two limitations were identified and explicitly carried forward into later runs:

- **Offset bound hit.** The warp fit's initial offset bound ($\pm$5 s) was hit during fitting (`bi[0]` = 1.099, at bound), producing a suboptimal initial warp and an inflated `warp_rmse` of 151 s in the sweep table. This is an artefact of an overly narrow search bound, not a failure of the underlying method; the planned fix, widening the bound to $\pm$100 s, was applied before subsequent sessions (R2–R4).
- **Trivial confusion matrix.** Because the offset-bound issue produced errors far larger than any reasonable true-positive tolerance, true/false positive/negative counts were uninformative (`tp/fp/fn = 0`) for this run and were not used to assess detection quality; MAE/RMSE on matched-pair times were used instead.

This phase's primary output (`sync_Camera1_*.json`) and lessons — particularly the value of ensemble confidence weighting and the danger of overly narrow optimisation bounds — carried forward conceptually into the ephys alignment work described in the next section, although the video and ephys alignment pipelines are methodologically independent.

3 Phase 1: Continuous-Feature Ephys Alignment

3.1 Problem Statement

A core requirement for analysing electrophysiology data from this task is aligning event timestamps between two independent clocks: the behavioural computer and the Neuropixels recording system (30 kHz AP sampling). These clocks drift apart over a session at roughly 1–6 ppm, amounting to 4–20 ms of accumulated error per hour. Where a hardware sync channel (the last AP channel, index 384) is connected and receiving TTL pulses from the behaviour computer, this alignment is trivial: match edges to behaviour events and fit a linear warp, yielding sub-millisecond precision.

The central problem addressed in this phase is that the hardware sync channel was not functional in the sessions of primary interest. Inspection of the sync channel across all 89 AP binary files in the dataset confirmed the channel was flat (constant value 64, a floating input with no TTL signal):

```
data = np.memmap(ap_bin, dtype=np.int16, mode='r', shape=(n_samp, 385))
sync = data[-500000:, 384]
np.unique(sync)  # -> [64]  (constant -- no TTL pulses)
```

With no hardware sync available in these sessions, the problem reduces to: find a neural or physical signal in the ephys recording that is reliably time-locked to a known behavioural event.

3.2 Pipeline Architecture

The alignment pipeline (`ephys_alignment_fusion.py`) was structured in two stages.

Stage 1 — Feature extraction and caching. The AP binary (typically 78 GB for a 56-minute session at 30 kHz $\times$ 385 channels) and LF binary (2.4 GB at 2500 Hz) are read once in overlapping 60-second chunks. Initial features extracted per chunk:

- **solenoid_artifact**: per-sample median across all 384 channels (common-mode signal), under the hypothesis that the water delivery solenoid produces a detectable electromagnetic transient. Z-scored absolute value; candidates via `find_peaks`.
- **solenoid_block**: variant computing the median over overlapping 32-channel depth blocks before taking the cross-block median, intended to reduce spatial noise relative to the global common-mode trace.
- **solenoid_derivative**: absolute differential of the global common-mode trace, intended to sharpen the onset edge.
- **mua_envelope**: bandpass 300–3000 Hz, rectify, smooth (25 ms Gaussian), downsample to 1 kHz.
- **lfp_deflection**: delta/theta/beta/gamma band features per depth block, PCA-projected, downsampled to $\sim$1170 Hz.
- **lick_band**: high-frequency LFP envelope (80–500 Hz) as a licking-artefact proxy.

All features were cached to `.npz` after extraction so subsequent analysis did not require re-reading the raw binaries.

Stage 2 — Event detection, warp fitting, and audit. For each behavioural reward event (code 2), the pipeline searched feature traces within a $\pm$500 ms window for the highest-prominence peak as the candidate ephys event. A linear warp (stretch + offset) was fitted by minimising RMSE between predicted and detected event times on an 80% training split, held out 20% for testing.

3.3 Extended Feature Engineering

When the initial solenoid-based features proved insufficient, an extended feature set (extract_extra_features.py extract_better_features.py) added ∼420 further features:

- **Band-specific block PCA** (lfp_block_<N>_<band>_pc{1,2,3}): per-depth-block PCA of bandpass-filtered LFP in delta, theta, beta, gamma bands; 19 blocks × 4 bands × 3 PCs = 228 features.
- **AP block MUA/HFO/gamma PCA** (ap_block_<N>_<band>_pc{1,2,3}): same structure on AP-band features; 57 additional features.
- **Reward Modulation Index, RMI** (rmi_block_<N>_<band>_pc{1,2,3}): the only supervised feature family. Per channel, a post/pre-reward log power ratio is computed across all reward events; the top-K most-modulated channels per depth block are selected and projected through PCA.
- **CAR-MUA** (car_<band>_block_<N>_pc{1,2,3}): common-average-referenced MUA. The per-timepoint median across all 384 channels is subtracted before bandpass filtering, eliminating common-mode artefact (notably the solenoid transient) before power estimation, so that subsequent PCA finds genuine spatial structure rather than artefact.
- **Spike Density Function, SDF** (sdf_block_<N>_pc{1,2,3}): threshold-crossing density — samples exceeding 4× the MAD-estimated noise floor, counted per millisecond bin, summed across channels in a depth block, Gaussian-smoothed. A more noise-robust MUA proxy than RMS envelope.
- **Phase-Amplitude Coupling, PAC** (pac_block_<N>, pac_tg_block_<N>): instantaneous mean-vector-length coupling between delta-phase/gamma-amplitude (and theta-phase/gamma-amplitude) per depth block, output at 10 Hz.

In total, the cache for a single session contained 440–520 features across all families.

3.4 Audit Results

Features were evaluated by two complementary metrics: an **F1-based audit** (plot_audit.py; precision/recall of find_peaks candidates at 2.5 SD prominence relative to reward events within ±500 ms) and **ERP visualisation** (feature_explorer.py, plot_individual_erps.py; trial-aligned trace averaging).

3.4.1 F1 Audit Findings

The highest F1 scores across all 440+ features were modest:

Table 4: Best-performing continuous features by F1 score.

Feature	F1	Hit rate	Med. error (ms)
pac_block_8	0.118	0.318	217
pac_block_9	0.114	0.298	170
rmi_block_7_theta_pc3	0.094	0.500	165
rmi_block_15_theta_pc3	0.093	0.530	194
rmi_block_3_theta_pc3	0.091	0.542	189

These F1 values (0.09–0.12) are very low; a feature suitable for alignment would require F1 > 0.5 and median error < 30 ms. The low values reflect a structural problem: **the features fire too frequently relative to reward events.** The best PAC feature generated ∼1,500 peaks per session against 336 rewards; with a 500 ms hit window, roughly 30% of peaks coincidentally fall near a reward even if the feature is entirely uncorrelated with it. Figure 3 shows this concretely for the `solenoid_block` feature: although its hit rate against rewards is high (0.96), its false-peak rate is equally high (0.95), with 13,055 total candidate peaks detected against only 336 reward events — the feature is sensitive but not remotely specific.

solenoid_block_full_session.png

Figure 3: Feature audit, `solenoid_block`, full session. **Top:** z-scored trace (grey) with detected peaks above threshold (gold dashed line), hits (blue dots, $n = 324$) and misses (red crosses, $n = 12$) relative to the 336 logged rewards. **Centre:** inter-event interval over the session, with the largest behavioural gap (146 s) highlighted. **Bottom:** per-event timing error (ms), median 178 ms (orange dashed line). Despite a high nominal hit rate (0.96), the false-peak rate (0.95) and $n = 13{,}055$ total candidate peaks confirm the feature is not specific to reward delivery, and the 178 ms median timing error is far too large for reliable per-trial alignment.

Importantly, low F1 does not indicate the features contain no neural signal — it indicates they are not *specific* enough to serve as a clock anchor. The distinction between alignment (requiring specificity) and neural-correlate analysis (requiring only consistency across trials) is central to interpreting this phase's results.

3.4.2 ERP Findings

ERP plots using the existing (poor) alignment showed flat mean traces for virtually all features. This was initially misattributed to poor feature quality; the correct diagnosis, confirmed by inspecting the sync JSON, was that the alignment itself had RMSE = 167 ms — larger than many neural response latencies — causing trial averaging to wash out any real signal:


```
{
  "warp": {"stretch": 1.000006, "offset": 0.165, "rmse": 0.167},
  "events": []
}
```


The empty `events` array confirmed that no reliable anchor points had been found; the warp was estimated purely from broad feature correlation, giving only an approximate offset with no per-trial precision. Figure 4 illustrates this directly for `car_gamma_block_2`: across all four available event types (Trial init/water, Reach, Pull, Inverse pull), the trial-averaged z-scored trace is essentially flat at zero, with no consistent event-locked deflection visible above the single-trial noise envelope.

erp_car_gamma_block_2.png

Figure 4: ERP audit, `car_gamma_block_2`, window 1500–2000 s. Each panel shows individual trials (grey) and the trial mean (black) aligned to a different behavioural event. **Top left:** Trial init / water ($n = 61$), with the median water-delivery latency (+221 ms) marked. **Top right:** Reach ($n = 61$). **Bottom left:** Pull ($n = 17$). **Bottom right:** Inverse pull, no events in this window. In all four conditions the mean trace is flat, consistent with the alignment RMSE (167 ms) being too coarse to resolve any genuine event-locked response.

A small number of features did show marginal ERP structure despite the poor alignment: `rmi_block_3_delta` showed a $z \approx 0.4$ bump at +100–200 ms in the Pull condition ($n = 17$ trials), consistent with delta-band modulation during action execution, but the effect was too weak and the trial count too low to draw conclusions. One exception of note is `car_mua_block_6_pc3`, shown in Figure 5: across 246 Pull trials this feature shows visible trial-to-trial structure (alternating red/blue bands in the trial-sorted heatmap) and a mean trace with small but repeated deflections around the event, distinguishing it from the flat traces typical of the other 439 features audited. This was the first hint that *some* genuine signal was present in the continuous data, even though it did not translate into a feature precise enough for alignment.

car_mua_block_6_pc3_Code_11.png

Figure 5: Trial-resolved ERP, car_mua_block_6_pc3, Pull event (code 11, $n = 246$ trials). **Top:** per-trial heatmap, trials on the y-axis, time from event on the x-axis, colour = z-score. Visible horizontal banding (clusters of consistently red or blue trials) indicates this PC captures slow, session-scale structure in addition to any event-locked component. **Bottom:** trial-averaged trace (±SEM, grey band); small repeated deflections are visible around the event but are not large enough relative to trial-to-trial variability to serve as a precise alignment anchor.

3.5 Root Cause Analysis

Three factors explain why the feature-based approach failed to achieve alignment precision:

1. **Solenoid artefact absent or undetectable.** The original design assumption — that the water-delivery solenoid would produce a large electromagnetic transient visible as a common-mode deflection across all 384 channels — did not hold in practice; the signal either did not reach the headstage or was shielded. The common-mode trace showed ∼1,849 peaks per session against 336 rewards, all attributable to movement, electrical noise, and licking rather than the solenoid valve.
2. **Neural features are not reward-specific.** MUA, LFP power, and PAC all respond simultaneously to movement, licking, arousal, and reward. In a joystick-pull task the animal is active throughout a large fraction of the session, so any feature responsive to general activity generates far more peaks than rewards, driving precision down regardless of recall.
3. **The 167 ms RMSE warp was itself an artefact of failed feature matching.** The warp's offset (∼165 ms) matched the known solenoid physical latency, but with no per-trial precision. Applying this warp and computing ERPs introduces ±167 ms of per-trial jitter, which washes out neural responses with latencies shorter than that jitter when averaged across trials.

3.6 Hardware Sync Channel Survey

Following the alignment failure, all 89 AP binary files were surveyed for sync-channel content:

```
sync_vals = np.unique(data[-500000:, 384])
# [64]      -- flat, no TTL (majority of sessions)
# [0, 64]   -- two-level signal indicating TTL pulses (minority)
```

Sessions with functional `[0, 64]` sync: Milka S2 imec0, Milka S3 imec0/imec1, Ferrero S2 imec1, and one additional Milka session. This confirmed the hardware sync was physically connected but not recording in the sessions of primary interest, while functional in some others, establishing a taxonomy of **ground-truth sessions** (hardware sync present) and **blind sessions** (alignment requires an alternative method). This taxonomy was later reused directly in Phase 2 to validate the spike-sorted method against ground truth (Section 4.5).

3.7 Transition Rationale: Spike-Sorted Anchor Units

The failure of feature-based alignment with 440+ continuous features motivated a fundamentally different approach: rather than searching for a continuous signal that peaks near reward events, search for an *individual neuron* that fires reliably at a fixed latency after reward delivery.

Table 5: Continuous feature vs. spike-sorted anchor unit.

Property	Continuous feature	Spike-sorted anchor unit
Events per session	1,000–16,000	200k–500k spikes total, matched per trial
Matching strategy	Global peak detection	Windowed per-trial search
Effect of baseline activity	High baseline → low precision	Baseline spikes within window are random; re
Clock drift sensitivity	High (warp from noisy matches)	Low (sigma-clipped fit over 100–300 pairs)
Expected RMSE	150–200 ms (observed)	5–20 ms (design target)

`anchor_unit_hunter.py` ranked all spike-sorted units in a session by reward-locked sharpness score (peak firing rate minus baseline in a 500 ms post-reward window) and trial reliability (fraction of trials with at least one spike near the peak latency). On the primary test session (Ferrero R1 imec0, 210 good units) this identified Unit 358 with sharpness = 21.4 Hz above baseline, peak latency = 215 ms, reliability = 80%, demonstrating that a suitable anchor unit existed in data that had been available all along in the Kilosort4 output.

3.8 Phase 1 Summary

Table 6: Alignment quality across all Phase 1 methods attempted.

Method	RMSE (ms)	Events matched	Notes
Solenoid common-mode (global)	177	0	Artefact not present
Solenoid derivative	182	0	Artefact not present
MUA envelope	173	0	Too many false peaks
LFP deflection	144	0	Best continuous feature; still unusable
All features fused	179	0	No improvement from fusion
Spike-sorted anchor (Unit 358)	**10.75**	**276 / 336**	**See Section 4**

The 15-fold reduction in RMSE from the best continuous feature (144 ms) to the spike-sorted anchor (10.75 ms) demonstrates that the alignment problem was solvable for this dataset, but required moving to the spike-sorted domain.

4 Phase 2: Spike-Sorted Anchor Alignment and Neural Circuit Analysis

4.1 Data

Eight probe recordings were analysed (Table 7). All probes used Neuropixels 1.0 (384 neural channels, 30 kHz AP sampling). Spike sorting used Kilosort4 (or the IBL sorter for some Milka imec1 sessions); manual curation in Phy labelled units *good* or *ExtRaGood*.

Table 7: Sessions and alignment statistics.

Animal	Session	Probe	Anchor unit	Pairs	RMSE (ms)
Ferrero	R1	imec0	Unit 358, 215 ms	276	10.75
Ferrero	R1	imec1	Unit 582, 305 ms	308	18.42
Ferrero	R4	imec0	Unit 515, 175 ms	115	13.57
Ferrero	R4	imec1	Unit 434, 175 ms	123	18.66
Milka	R1	imec1	Unit 27, 25 ms	127	16.44
Milka	R4	imec0	Unit 102, 55 ms	210	6.12
Milka	R4	imec1	Unit 283, -95 ms	211	8.77
Milka	R2	imec0	3-unit fusion	140	1.62

4.2 Anchor Unit Selection

`anchor_unit_hunter.py` ranked all good units by:

$$\text{score} = \text{FR}_{\text{peak}} - \text{FR}_{\text{baseline}} \quad [\text{Hz}]$$

computed in 10 ms bins over a ±500 ms window around reward events. Trial reliability (fraction of trials with at least one spike near the peak latency bin) was used as a secondary filter; units with reliability $< 60\%$ were not selected.

4.3 Spike-Based Clock Warp

`spike_sync_generator.py` procedure:

1. Zero-anchor behaviour log to code 1000 (ephys start).
2. For each reward event t_i^{bhv}, search the anchor spike train in $[t_i^{\text{bhv}} + \hat{\lambda} - w,\ t_i^{\text{bhv}} + \hat{\lambda} + w]$ ($\hat{\lambda}$: peak latency; $w = 50$ ms).
3. Match to nearest spike t_i^{ephys}.
4. Reject outliers: 3-σ iterative clipping.
5. Least-squares linear warp: $t^{\text{ephys}} = s \cdot t^{\text{bhv}} + \delta$.

Result saved as `spike_sync.json` (s, δ, RMSE, matched pairs).

4.4 Multi-Anchor Fusion

For Milka R2 imec0 (single-anchor match rate 68%), three units were fused with `multi_anchor_sync.py`. Per trial, the per-trial ephys estimate is the weighted median across contributing units (weights $\propto$ sharpness score). RMSE dropped from 2.14 ms (best single unit) to **1.62 ms**.

4.5 Independent Validation Against Hardware Sync

Where the Phase 1 sync-channel survey identified a functional hardware TTL signal, `validate_spike_sync.py` was used to independently check the spike-anchor warp against ground truth: hardware edge times are matched to reward events by ISI pattern, fitted to their own linear warp, and the two warps' predictions are compared directly. For Ferrero S1 imec0, the hardware warp ($s = 0.99999391$) and the spike warp ($s = 1.00000034$) agreed to within 6.98 ms RMSE on predicted event time across 333 matched pairs after correcting for the $\sim$112 s constant time-base offset between the two clocks' absolute zero points — well within the 20 ms tolerance used elsewhere in this report, and consistent with the 10–19 ms RMSE range reported for the spike-anchor method on sessions without independent ground truth. This provides direct, independent confirmation that the spike-sorted anchor method achieves the precision claimed in Table 7, rather than relying solely on the internal consistency of the sigma-clipped fit.

4.6 Neural Circuit Analysis Methods

Behaviour event times are warped: $t_i^{\text{ephys}} = s \cdot t_i^{\text{bhv}} + \delta$. For each good unit a PSTH is computed in 20 ms bins over ± 1 s. Z-scoring uses the per-trial baseline distribution:

$$Z(t) = \frac{\text{FR}(t) - \mu_{\text{base}}}{\sigma_{\text{base}}}$$

where μ_{base} and σ_{base} are the mean and SD of per-trial baseline firing rates (-1 to -0.1 s).

Classification at $|Z| > 2.0$:

- **MOTOR**: peak $Z > 2.0$ pre-event (-1 to 0 s).
- **REWARD**: peak $Z > 2.0$ post-event (0 to $+1$ s).
- **SUPPRESSED**: trough $Z < -2.0$ post-event.
- **UNRESPONSIVE**: none of the above.

For the neural manifold, the smoothed Z-scored PSTH matrix $\mathbf{A} \in \mathbb{R}^{T \times N}$ (time bins $\times$ responsive units) was projected to 3 dimensions with PCA.

4.7 Alignment Results

Figure 6 shows alignment diagnostics for Ferrero R1 imec0 (276 matched pairs, RMSE = 10.75 ms). The stretch $s \approx 1.000002$ (≈ 2 ppm clock rate difference) is constant over the session (flat running median of residuals), confirming that a single linear warp is sufficient. The median per-trial matching error is 3.9 ms.

1_warp_analysis.png

Figure 6: Warp diagnostic, Ferrero R1 imec0. **Left:** Behaviour vs. ephys time for 276 matched reward events; red line: fitted warp ($s = 1.000002$, $\delta = 211$ ms). **Centre:** Residuals over session; running median (blue) is flat, confirming no nonlinear drift. **Right:** Per-trial absolute matching error; median 3.9 ms, RMSE 10.75 ms.

Figure 7 shows the full session event timeline. A cluster of inverse-pull events (trials 50–90) identifies a period of task disengagement that should be excluded from learning-related analyses.

5_session_timeline.png

Figure 7: Session timeline, Ferrero R1 imec0. **Top:** All coded events vs. session time. **Bottom:** Inter-trial interval (mean 14.7 s); one outlier ($\approx$ 1100 s) marks a break.

4.7.1 Anchor Unit

Figure 8 shows Unit 358. The raster confirms reliable firing at 215 ms post-pull (80% trial reliability). ISI violations (7.92%) are elevated but do not affect the warp because sigma-clipping removes outlier matches before fitting.

4_anchor_unit.png

Figure 8: Anchor Unit 358, Ferrero R1 imec0. **Top:** Raster across 246 pull trials. Green line: 215 ms latency window. **Middle:** Z-scored PSTH. **Bottom:** ISI distribution (7.92% refractory violations).

4.8 Neural Circuit Results

4.8.1 Unit Classification

Across 210 good units (Ferrero R1 imec0, RMSE 10.75 ms):

Class	n	%	ExtRaGood
MOTOR	46	21.9	14
REWARD	119	56.7	32
SUPPRESSED	36	17.1	14
UNRESPONSIVE	9	4.3	6

In the 1.62 ms-RMSE session (Milka R2 imec0, 109 units): 17 Motor (15.6%), 73 Reward (67.0%), 2 Suppressed (1.8%), 17 Unresponsive (15.6%). The higher Reward fraction and lower Suppressed fraction likely reflect differences in probe placement rather than alignment quality.

2_classification.png

Figure 9: Unit classification, Ferrero R1 imec0 (n = 210). **Left:** Class proportions. **Right:** Baseline firing rate by class (log scale). Motor and Suppressed units have higher baseline rates than Reward units, consistent with interneuron vs. output neuron subtypes in striatum.

4.8.2 Population Timing Structure

Figure 10 shows the peak latency distribution. Motor units cluster in $[-600, 0]$ ms; Reward units span $[0, +1000]$ ms with a secondary peak near +600 ms (possibly corresponding to water contact and consummatory licking onset). The Z-score vs. latency scatter confirms that ExtRaGood units (stars) are distributed across both pre- and post-event windows.

3_latency_map.png

Figure 10: Population timing map, Ferrero R1 imec0, code 11. **Left:** Peak latency histogram by class. **Right:** Peak Z-score vs. latency; ExtRaGood units marked as stars.

4.8.3 Individual Unit Examples

Unit_0076_code11.png

(a) Unit 76, MOTOR.
$Z = 2.4$ at -10 ms.

Unit_0077_code11.png

(b) Unit 77, SUPPRESSED (ExtRaGood).
$Z = -2.7$ at $+10$ ms.

Unit_0025_code11.png

(c) Unit 25, REWARD.
$Z = 2.9$ at $+950$ ms.

Figure 11: Representative units, Ferrero R1 imec0, Pull (code 11). Each panel: raster (top) and Z-scored PSTH (bottom). Dashed lines: ± 2 SD classification threshold.

4.8.4 Population Heatmap

Figure 12 shows the population heatmap for Ferrero R1 imec0. Units sorted by peak latency reveal a clear temporal cascade: Motor units (bottom) are active before $t = 0$ and suppressed after; Reward units (middle) show post-event activation spreading from 0 to $+1$ s; Suppressed units (upper) show a sustained post-pull decrease.

population_heatmap.png

Figure 12: Population activity heatmap, Ferrero R1 imec0 ($n = 210$, code 11). Units sorted by peak latency. Colour: Z-score relative to baseline. The temporal cascade from motor preparation (red, bottom, pre-event) through reward processing (red, middle, post-event) is consistent with cortico-striatal circuit organisation during goal-directed behaviour.

4.8.5 Neural State-Space Trajectory

Figure 13 shows the 3-dimensional neural manifold for Ferrero R1 imec0. PCA applied to the $T \times N$ PSTH matrix of 201 responsive units; PC1: 41.5% variance, PC2: 14.5%, PC3: 7.9% (cumulative 63.9%). The pre-event (blue) and post-event (red) trajectories occupy distinct regions of state space, with the pull event (gold star, $t = 0$) marking a sharp inflection. The trajectory does not return to the baseline region within +1 s, indicating a sustained population state change following reward delivery.

neural_manifold_code11.png

Figure 13: Neural state-space trajectory, Ferrero R1 imec0 ($n = 201$ responsive units, code 11). Axes: first three PCs of the population PSTH matrix. Colour: time from event (-1 to $+1$ s). Gold star: $t = 0$ (Pull). The pre- and post-event trajectories are geometrically separated, indicating a qualitative change in population state at movement onset.

5 Cross-Phase Discussion

5.1 Why Each Phase Succeeded or Failed

The three phases of this project illustrate a single recurring principle: **alignment precision requires a signal that is specific to the target event, not merely correlated with it.** The video phase succeeded qualitatively (138/138 trials aligned) but retained second-scale imprecision because its ensemble detector, however well-tuned, was ultimately matching a noisy visual proxy under a narrow and ultimately limiting optimisation bound. The ephys feature phase failed outright because every one of 440+ candidate continuous signals responded to a broad mixture of behavioural and physiological events, not reward delivery specifically, so no amount of feature engineering could substitute for genuine specificity. The spike-sorted phase succeeded because individual neurons, unlike population-level continuous signals, can be both highly reward-specific (in their peak response) and quiet at other times (their baseline firing in the matching window is effectively random with respect to reward timing), which is exactly the property a clock-anchor signal needs.

5.2 Alignment Quality Across Methods

Table 8: Alignment precision achieved by each phase's primary method.

Phase	Method	Typical RMSE
0	Video ensemble (PC1 + template + stability)	∼3.7 s (post-fit), ∼bound-limited
1	Best continuous ephys feature (LFP deflection)	144 ms
1	All features fused	179 ms (no improvement)
2	Spike-sorted single anchor unit	1.6–18.7 ms
2	Spike-sorted multi-anchor fusion	1.62 ms (best)

The spike-sorted approach achieved RMSE of 1.6–18.7 ms across all eight probe recordings, a 10–100-fold improvement over the continuous-feature baseline (140–200 ms) and several orders of magnitude finer than the second-scale precision of the video alignment. Multi-anchor fusion reached 1.62 ms — the practical floor given Kilosort's ∼1 ms sorting jitter. Sessions with higher RMSE (18 ms) used anchor units with lower reliability or higher baseline variability; even at 18 ms, this remains well below the 20 ms PSTH bin size used in circuit analysis and introduces negligible smearing. Independent validation against the hardware sync channel (Section 4.5) confirmed this precision is not an artefact of internal self-consistency in the sigma-clipped fit.

5.3 Biological Interpretation

The high responsive fraction (95.7% at $|Z| > 2.0$ in the primary test session) reflects the recording locations in striatum and motor/somatosensory cortex. A threshold of 3.0 SD would be more appropriate for formal statistical tests. The broad Reward latency distribution (0 to +1 s, secondary peak at ∼600 ms) suggests heterogeneous reward signals: action outcome (∼200 ms), water contact, and consummatory licking. The neural manifold's failure to return to baseline within +1 s is consistent with sustained post-reward activity in dorsal striatum reported in prior rodent electrophysiology studies.

5.4 Limitations

- **Single-session focus.** The detailed circuit analyses in Section 4 are from one session (Ferrero R1 imec0). Cross-session and cross-animal replication is required before biological conclusions can be drawn.

- **ISI violations.** Unit 358 has 7.92% refractory-period violations, suggesting possible multi-unit contamination. This does not affect alignment quality, since sigma-clipping is robust to a moderate rate of spurious spikes.
- **Classification threshold.** $|Z| > 2.0$ was chosen empirically for unit classification. A permutation test would be preferable for formal statistical claims.
- **Inverse-pull period.** Trials 50–90 (Ferrero R1) should be excluded from any motivationally sensitive analysis, as they reflect a period of task disengagement rather than goal-directed behaviour.
- **Video alignment bound.** The Phase 0 offset bound (± 5 s) constrained the initial video warp fit; later sessions used a widened ± 100 s bound, so Milka R1's reported RMSE should not be taken as representative of the method's best achievable precision.

6 Conclusion

This project addressed behavioural-to-recording-system alignment across three independent signal domains. Video-based alignment (Phase 0) established that ensemble confidence-weighted detection could align 100% of trials at second-scale precision, while also surfacing the importance of adequately wide optimisation bounds for warp fitting. Continuous-feature ephys alignment (Phase 1) tested 440+ engineered features across solenoid-artefact, MUA, LFP-band, PAC, and reward-modulation-index families, and established conclusively that none were specific enough to reward delivery to support precise alignment — the best continuous feature achieved only 144 ms RMSE, and naive feature fusion did not improve on this. This motivated the transition to spike-sorted anchor units (Phase 2), which achieved RMSE of 1.6–18.7 ms across eight probe recordings, a 10–100-fold improvement, independently validated against the hardware TTL sync channel where available.

With alignment secured at the precision required for sub-PSTH-bin analysis, the same spike-sorted data reveals a clear functional organisation in the test session: motor units active before the pull, reward units responding in the 0–1 s post-pull window, and a population state-space trajectory with geometrically separated pre- and post-event regions, consistent with cortico-striatal circuit organisation during goal-directed behaviour.

The pipeline (`anchor_unit_hunter.py`, `spike_sync_generator.py`, `multi_anchor_sync.py`, `validate_spike_s` `plot_neural_circuits.py`) is fully automated and session-agnostic, and is ready for application across all eight sessions and both animals. The next phase will examine how Motor/Reward cell proportions and latency distributions vary across sessions (learning) and probe locations (anatomy).